

Internal Round Prototype Scope

Deadline: Sept 21 | Deliverable: Basic working hardware prototype + full PPT

1. Golden Rule

Build the smallest thing that works and is verifiable live in the room. Everything else goes on a roadmap slide in the PPT, clearly labeled as future work — not demoed as built.

2. What We ARE Building for Sept 21

One feature, fully working:

IMU device tracks barbell motion during a lift → sends data over BLE → app calculates and displays **rep velocity in real time**, plotted as a **velocity-per-rep graph across a set**.

That's it. No load recommendation engine yet (that's Phase 1, PPT-only for now, described not demoed — unless time genuinely allows after this core works). No camera. No rehab logic. No dashboards.

Why this scope and not more

- It's genuinely achievable in ~11 days by a small team
 - It's **live-demoable and verifiable** — the strongest possible thing to show judges (they can watch it happen in real time, not a pre-recorded video)
 - It proves the hardest technical piece (motion sensing + BLE + real-time velocity calc) actually works, which is the foundation everything else depends on
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3. Task Breakdown

3.1 Firmware / Hardware (Owner: _____)

- ☐ Set up Seeed XIAO nRF52 + BMI270 (SPI) breakout
- ☐ Read accelerometer + gyro data at a consistent sample rate (target $\geq 100\text{Hz}$ for usable velocity resolution)
- ☐ Implement basic velocity estimation: integrate acceleration over time per rep
- ☐ Implement a simple **Zero-Velocity Update (ZUPT)**: detect the static pause at top/bottom of a rep (near-zero acceleration + gyro) and reset integrated velocity to zero there — this is what keeps drift from making the numbers meaningless
- ☐ Stream data over **BLE** to the app (define a simple data packet format now — timestamp, accel x/y/z, gyro x/y/z)
- ☐ Add button logic: press to start a set, press to end a set
- ☐ Mount sensor in a small magnetic housing (3D print or off-the-shelf enclosure is fine for this stage — doesn't need to be final design)
- ☐ Battery: confirm runtime is enough for a demo session (30–60 min is plenty for Sept 21)

3.2 App / Software (Owner: _____)

- ☐ BLE receiver (mobile app or simple laptop/web app — pick whichever the team can build fastest and reliably for a live demo)
- ☐ Parse incoming IMU stream, run velocity calculation per rep (can mirror or replace firmware-side calc — decide with firmware owner)
- ☐ Real-time display: current rep's mean velocity + peak velocity
- ☐ After a set: simple **bar graph — velocity per rep across the set** (this single graph is your "wow" moment — it visually shows fatigue as bars get shorter across the set)
- ☐ Basic, clean UI. Doesn't need to be beautiful — needs to be **legible from a few feet away** when judges are watching

3.3 PPT (Owner: _____, but everyone reviews)

- ☐ Follow structure in Section 5 below
- ☐ Use the elevator pitch and story from `01_SIH2026_Idea_Vision.md`
- ☐ Clearly mark what's built (Phase 0) vs. roadmap (Phase 1-3) — do not blur this line, judges specifically probe for it
- ☐ Include the competitive landscape table (Section 6 of the vision doc) — shows market awareness
- ☐ Include 1-2 photos/screen recording of the actual working prototype demo as backup in case live demo has technical issues

3.4 Prototype Demo Rehearsal (Owner: whole team)

- ☐ Rehearse the live demo at least 3 times before Sept 21, including recovering from a failure (BLE disconnect, low battery, etc.)
 - ☐ Have a **backup video recording** of a successful run, ready to show instantly if live demo fails
 - ☐ Decide who does the physical lift during demo (should be comfortable, controlled reps — doesn't need to be heavy weight, just consistent)
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4. What We Are NOT Building Yet (and why)

Feature	Why it's excluded from this round
Load-to-RPE recommendation engine	Needs the velocity tracking foundation to be solid first — build after Sept 21, for main PPT/next phase
Camera-based asymmetry detection	Separate technical stack (CV), real risk of splitting team focus and delivering two half-working things instead of one solid thing
Rehab decision-support system	Content-heavy (needs the coach's domain logic mapped out properly), not a quick build — do this properly for Phase 2, not rushed
Dashboards, athlete/coach accounts, history	Infrastructure work with no demo payoff at this stage

If there's spare time after the core velocity tracker is rock-solid, the **next best use of remaining days** is a simple, hard-coded mockup slide/screen showing what the load recommendation UI *will* look like — not a working backend, just a believable mockup for the PPT. Do not let anyone attempt to build the rehab logic or camera feature "just in case" — it will eat time the core feature needs.

5. Suggested PPT Structure (Internal Round)

1. **Problem** — injury risk from untracked training load; commercial VBT devices priced out of reach in India
2. **Our solution** — the one-line pitch + the 4-stage loop diagram (Track → Autoregulate → Detect → Recover)
3. **What's built today** — the working velocity tracker, live demo, graph screenshot
4. **How it works (technical)** — IMU + BLE + ZUPT drift correction (shows technical depth)
5. **Competitive landscape** — table showing pricing/gap vs. GymAware, Output Sports, Vmax Pro, Stance, Lift.app.ai
6. **Roadmap** — Phase 1 (load engine) → Phase 2 (rehab logic, dashboards) → Phase 3 (camera cross-validation)
7. **Team & why us** — include the S&C coaching background explicitly; it's a real credibility asset, use it
8. **Impact** — accessibility (cost), injury reduction potential, who benefits (coaches, athletes, academies)

6. Timeline (11 days)

Days	Focus
Day 1-2	Finalize BLE data packet format, get raw sensor streaming end-to-end (garbage data is fine at this stage — just prove the pipe works)
Day 3-5	Velocity calculation + ZUPT drift correction on firmware/app side
Day 6-7	App UI: real-time display + post-set graph
Day 8	Enclosure/mounting, battery check, first full rehearsal
Day 9	PPT draft complete, second rehearsal
Day 10	Buffer for fixing whatever broke in rehearsal (something always does)
Day 11 (Sept 21)	Final rehearsal morning-of, present

Buffer days matter more than feature days at this stage — protect Day 10.